



El-Sewedy University of Technology

Faculty of Engineering Technology

Graduation Project II



LeafSense – AI-Powered Plant Disease Detection System

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Abstract

Plant diseases are among the leading causes of crop losses worldwide, significantly affecting agricultural productivity, food security, and farmers' income. Traditional disease diagnosis methods often rely on expert inspection and laboratory testing, which can be costly, time-consuming, and inaccessible for many farmers, particularly in rural areas. To address this challenge, this project developed LeafSense, an AI-powered mobile application for automated plant disease detection and diagnosis.

The proposed system utilizes deep learning and computer vision techniques to identify plant diseases from leaf images captured using a smartphone. The solution integrates a Flutter-based mobile application, a FastAPI backend server, Firebase cloud services, and a deep learning classification model based on EfficientNetV2-S with transfer learning and fine-tuning techniques.

The model was trained and evaluated using a dataset consisting of 54,305 images distributed across 38 classes representing plant diseases and healthy plant conditions. The dataset was divided into training, validation, and testing subsets using a 70%-15%-15% split. Experimental evaluation demonstrated excellent performance, achieving a classification accuracy of 99.77%, an F1-score of 99.66%, and a Test-Time Augmentation (TTA) accuracy of 99.78%.

The developed application enables users to capture or upload plant images, receive disease predictions in real time, and access treatment recommendations and preventive guidance. The system provides an accessible and user-friendly solution for farmers, agricultural engineers, researchers, and plant enthusiasts.

The results demonstrate that deep learning-based disease diagnosis can provide highly accurate and scalable agricultural decision support. LeafSense contributes to the modernization of agriculture by offering an intelligent, affordable, and practical tool for early plant disease detection and management.

Keywords: Plant Disease Detection, Deep Learning, EfficientNetV2-S, Computer Vision, Agriculture, Mobile Application

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List of Abbreviations

Abbreviation	Meaning
AI	Artificial Intelligence
CNN	Convolutional Neural Network
ML	Machine Learning
DL	Deep Learning
API	Application Programming Interface
TTA	Test-Time Augmentation
UI	User Interface
REST	Representational State Transfer
JSON	JavaScript Object Notation
Firebase	Google Cloud Backend Platform

Chapter 1: Introduction

1.1 Background and Context

Agriculture plays a vital role in ensuring food security, supporting economic development, and sustaining livelihoods across the world. In countries such as Egypt, agriculture represents a critical sector that contributes significantly to employment and national economic growth. However, agricultural productivity is continuously threatened by plant diseases, which can spread rapidly and cause severe crop damage if not identified and treated at an early stage.

Traditionally, plant disease diagnosis relies on manual inspection performed by experienced agricultural experts. While effective, this approach can be expensive, time-consuming, and difficult to access for farmers in remote and rural areas. Delays in diagnosis often result in reduced crop quality, lower yields, increased treatment costs, and substantial financial losses.

Recent advances in Artificial Intelligence (AI), Machine Learning (ML), and Computer Vision have created new opportunities for automated agricultural solutions. Deep learning models, particularly Convolutional Neural Networks (CNNs), have demonstrated remarkable success in image classification tasks and have been increasingly applied to plant disease detection. These technologies allow farmers to obtain rapid and accurate diagnoses using ordinary smartphone cameras without requiring specialized equipment.

LeafSense is an AI-powered mobile application designed to leverage these technological advancements to support farmers and agricultural professionals. By combining deep learning, mobile computing, and cloud-based services, the system provides an intelligent platform capable of detecting plant diseases and offering treatment recommendations in real time. The project focuses on improving accessibility, reducing diagnosis time, and enhancing agricultural productivity through modern AI-driven solutions.

1.2 Problem Statement

Plant diseases are responsible for significant agricultural losses worldwide and represent a major challenge for farmers, agricultural organizations, and food supply chains. Early disease detection is critical for preventing disease spread and minimizing crop damage; however, traditional diagnosis methods depend heavily on expert knowledge and manual inspection. In many rural and developing regions, farmers lack immediate access to agricultural specialists, resulting in delayed identification and treatment of plant diseases.

The inability to rapidly diagnose plant diseases often leads to reduced crop yields, increased pesticide usage, higher production costs, and economic losses. Existing solutions may also require expensive equipment, specialized training, or internet-dependent services that are not always available in agricultural environments. Therefore, there is a need for an intelligent, affordable, and scalable solution capable of automatically identifying plant diseases from leaf images while providing actionable recommendations to users. LeafSense addresses this challenge through a deep learning-powered mobile platform that enables fast and accurate disease diagnosis using smartphone technology.

1.3 Motivation and Justification

The growing global demand for food, coupled with increasing environmental and agricultural challenges, highlights the importance of adopting intelligent farming technologies. According to reports from the Food and Agriculture Organization (FAO), plant diseases contribute to substantial losses in global food production annually. These losses affect not only farmers' incomes but also national food security and economic stability.

At the same time, advances in artificial intelligence have made high-performance image classification systems more accessible and affordable than ever before. The widespread availability of smartphones creates an opportunity to deliver AI-powered agricultural assistance directly to farmers. This project was motivated by the need to bridge the gap between advanced AI technologies and practical agricultural applications, particularly within Egypt and the MENA region.

1.4 Project Objectives

- Objective 1: Design and develop an AI-powered mobile application capable of detecting and classifying plant diseases from leaf images.
- Objective 2: Implement a deep learning model based on EfficientNetV2-S to achieve high classification accuracy across multiple plant disease categories.
- Objective 3: Develop a complete mobile-to-cloud workflow integrating Flutter, FastAPI, TensorFlow, and Firebase services.
- Objective 4: Evaluate the performance of the proposed system using standard machine learning metrics including Accuracy and F1-Score.
- Objective 5: Provide users with disease information, treatment recommendations, and preventive agricultural guidance.

1.5 Scope and Limitations

The scope of this project includes the development of a software-based mobile application for automated plant disease classification using image analysis and deep learning techniques. The system supports disease recognition across 38 plant health classes using a dataset containing 54,305 images. The project covers mobile application development, backend API implementation, model training, evaluation, and deployment.

The project does not include IoT sensor integration, automated irrigation systems, weather forecasting, or real-time environmental monitoring. Furthermore, the model performance is limited by the diversity and quality of the available training dataset and may require retraining to support additional crops or disease categories.

1.6 Project Outcome and Deliverables

The LeafSense project resulted in the development of a complete AI-powered plant disease detection system that integrates deep learning, mobile technologies, and cloud-based services. The project deliverables include both software components and supporting documentation, ensuring that the system is fully functional, well-documented, and ready for demonstration and future development. The main outcomes and deliverables of the project are as follows:

- Fully functional AI-powered mobile application.
- Trained EfficientNetV2-S disease classification model.
- FastAPI backend service for prediction and report generation.
- Dataset preparation and model evaluation reports.
- Complete source code repository.
- Technical documentation and final project report.
- Demonstration video showcasing system functionality.

1.7 Document Organization

This document is organized into eight chapters. Chapter 2 presents the theoretical background and reviews existing literature related to plant disease detection and deep learning applications in agriculture. Chapter 3 evaluates the technical, economic, operational, and risk feasibility of the proposed system. Chapter 4 discusses the system design and technology selection process. Chapter 5 explains the methodology, architecture, and workflow used during development. Chapter 6 describes the implementation details of the software components and AI model. Chapter 7 presents the testing procedures, experimental results, and performance evaluation. Finally, Chapter 8 summarizes the project achievements, discusses limitations, and outlines future work opportunities.

Chapter 2: Literature Review

2.1 Theoretical Background

Plant disease detection has become one of the most important research areas in modern precision agriculture. Early identification of plant diseases enables farmers to apply corrective actions before infections spread, reducing crop losses and improving agricultural productivity. Traditionally, disease diagnosis has relied on visual inspection performed by agricultural experts. While effective, manual diagnosis is often subjective, time-consuming, and difficult to scale across large agricultural regions.

Recent advances in Artificial Intelligence (AI) and Machine Learning (ML) have enabled the development of automated disease diagnosis systems based on image analysis. Among these technologies, Deep Learning has emerged as the dominant approach due to its ability to learn complex visual patterns directly from image data without requiring manual feature engineering.

Convolutional Neural Networks (CNNs) are specialized deep learning architectures designed for image processing tasks. CNNs automatically extract hierarchical image features through convolutional, pooling, and fully connected layers. Lower layers learn basic visual features such as edges and textures, while deeper layers learn more complex patterns related to disease symptoms, lesions, discoloration, and leaf deformities.

Transfer Learning is another important technique used in plant disease detection. Instead of training a model from scratch, a pre-trained model is fine-tuned on a target dataset. This approach significantly reduces training time, improves generalization performance, and enables high accuracy even when domain-specific datasets are limited.

The PlantVillage dataset has become one of the most widely used benchmarks for plant disease classification research. It contains thousands of labeled images covering multiple crops and disease categories, making it suitable for training and evaluating deep learning models. Modern architectures such as ResNet, MobileNet, EfficientNet, and EfficientNetV2 have demonstrated exceptional performance on this dataset.

In addition to disease classification, modern agricultural applications increasingly integrate mobile computing, cloud services, and RESTful APIs to deliver intelligent diagnostic tools directly to farmers. Such systems enable real-time image analysis using smartphones, providing practical and scalable solutions for agricultural communities.

2.2 Related Work

Deep Learning-Based Plant Disease Detection

One of the earliest large-scale studies on plant disease classification using deep learning was conducted by Mohanty et al. [1]. The authors utilized the PlantVillage dataset to train Convolutional Neural Network (CNN) models capable of classifying healthy and diseased plants across multiple crop species. Their results demonstrated that deep learning techniques could achieve classification accuracies exceeding 99% under controlled laboratory conditions.

Ferentinos [3] evaluated several state-of-the-art CNN architectures for plant disease diagnosis, including AlexNet, VGG, and GoogLeNet. The study compared the performance of these models across numerous plant disease categories and showed that deep learning approaches significantly outperformed traditional machine learning methods, achieving classification accuracies above 99%.

The PlantVillage dataset itself was introduced by Hughes and Salathé [2], providing thousands of labeled plant images covering various crops and diseases. Since its publication, the dataset has become one of the most widely used benchmarks in agricultural computer vision research and has enabled significant advancements in AI-based plant disease detection.

Mobile and Real-Time Agricultural Diagnosis Systems

Ramcharan et al. [4] proposed a deep learning-based cassava disease detection system that could operate on mobile devices. Their approach enabled farmers to diagnose diseases directly in the field using smartphone cameras, demonstrating the practicality of integrating artificial intelligence into everyday agricultural activities.

In addition to academic research, several commercial solutions have emerged, such as PlantVillage Nuru and other AI-powered agricultural assistants. These systems provide disease diagnosis and treatment recommendations through mobile platforms. However, many of these solutions focus on specific crops or geographic regions, which may limit their applicability in different agricultural environments.

The reviewed studies collectively demonstrate the effectiveness of deep learning for plant disease detection. Nevertheless, challenges remain regarding real-world deployment, scalability, and adaptation to local crop varieties, motivating the development of the proposed LeafSense system.

Transfer Learning Approaches

Recent research has increasingly focused on transfer learning techniques due to their ability to improve classification accuracy while reducing computational requirements. Pre-trained models such as MobileNet, ResNet, and EfficientNet have been successfully adapted to agricultural datasets. Transfer learning enables models to leverage features learned from large image datasets, resulting in faster convergence and improved performance.

EfficientNet-based architectures have attracted particular attention due to their balance between computational efficiency and predictive performance. The EfficientNetV2 family further improves training speed and model scalability, making it suitable for deployment in cloud-connected mobile applications.

2.4 Limitations of Existing Systems

Despite significant progress in AI-based plant disease detection, several limitations have been reported in the literature [1]–[4]. Many existing systems face challenges that affect their practical deployment and adoption in real agricultural environments. These limitations include:

- Limited support for local crop varieties and region-specific diseases [1], [4].
- Insufficient adaptation to different environmental and agricultural conditions [1], [3].
- Dependence on high computational resources for model training and deployment [3].
- Lack of integration with user-friendly and accessible mobile applications [4].
- Limited availability of treatment recommendations and preventive guidance for farmers [4].
- Reduced performance when deployed outside controlled laboratory environments, compared to benchmark datasets such as PlantVillage [1], [2].

These limitations highlight the need for scalable, accurate, and region-specific solutions capable of supporting practical agricultural applications. The proposed LeafSense system aims to address several of these challenges through the integration of a high-performance deep learning model, mobile deployment, and treatment recommendation features.

2.3 Comparative Analysis

To better understand the current state of AI-based plant disease detection systems, a comparative analysis was conducted between representative studies from the literature and the proposed LeafSense system. The comparison focuses on the application domain, key achievements, and limitations of each approach. This analysis highlights the strengths of the proposed system and demonstrates how it addresses some of the limitations identified in previous research and existing commercial solutions.

Table 2.1 summarizes the comparison between representative plant disease detection approaches and the proposed LeafSense system.

Reference / Method	Application	Key Result	Limitation
Mohanty et al. [1]	Plant disease classification using CNNs	Accuracy above 99% on PlantVillage	Performance decreases in field conditions
Hughes & Salathé [2]	PlantVillage Dataset	Large open-access agricultural dataset	Images collected mainly in controlled environments
Ferentinos [3]	Deep CNN disease diagnosis	Accuracy exceeding 99%	High computational requirements
Ramcharan et al. [4]	Cassava disease detection	Mobile-based disease diagnosis	Focused on cassava crops only
Commercial AI Apps	Agricultural diagnosis assistance	Practical deployment and user accessibility	Limited regional customization
LeafSense (Proposed System)	Mobile disease detection for Egyptian agriculture	EfficientNetV2-S, 99.77% Accuracy, treatment recommendations, mobile deployment	Limited to supported disease classes

2.4 Research Gap and Contribution

The reviewed literature demonstrates that deep learning has achieved remarkable success in plant disease classification. However, many existing studies focus primarily on model development and laboratory evaluation rather than complete end-user solutions. Additionally, several systems are optimized for specific crops or regions and do not adequately address the needs of farmers in Egypt and the MENA region.

Another limitation is that many research works provide disease predictions without integrating practical agricultural guidance, mobile usability, and cloud-based deployment into a unified platform. Furthermore, several solutions require specialized hardware or are not designed for everyday use by non-technical users.

LeafSense addresses these gaps by providing a complete AI-powered agricultural assistance platform that combines high-accuracy disease classification, mobile accessibility, cloud deployment, treatment recommendations, and user-friendly interaction within a single system. The project utilizes an EfficientNetV2-S model trained on 54,305 images across 38 classes and achieves an accuracy of 99.77%, demonstrating the effectiveness of deep learning for practical agricultural applications.

The main contributions of this project can be summarized as follows:

- Development of a complete mobile-based plant disease detection platform.
- Integration of EfficientNetV2-S transfer learning for high-accuracy classification.
- Support for 38 disease and healthy plant classes.
- Delivery of treatment recommendations and preventive guidance.
- Cloud-based deployment using FastAPI and Firebase services.
- Adaptation of the solution to agricultural requirements within Egypt and the MENA region.

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Chapter 3: Feasibility Study

3.1 Technical Feasibility

The successful implementation of LeafSense depends on the availability of suitable technologies, development tools, computational resources, and technical expertise. An analysis of these factors indicates that the project is technically feasible and can be developed using currently available technologies.

From a software perspective, the project utilizes widely adopted technologies including Python, TensorFlow, Flutter, FastAPI, and Firebase services. These technologies provide extensive documentation, active community support, and proven reliability for machine learning and mobile application development.

The artificial intelligence component is based on the EfficientNetV2-S architecture, a state-of-the-art deep learning model known for its high accuracy and computational efficiency. The availability of transfer learning techniques further reduces training complexity and development time.

The required dataset resources are also readily available. The PlantVillage dataset provides a large collection of labeled plant disease images, while additional Egyptian plant images improve the model's applicability to local agricultural conditions.

The development team possesses the necessary knowledge in artificial intelligence, mobile application development, computer vision, cloud services, and software engineering, making implementation achievable within the project timeframe.

Furthermore, integration between the mobile application, backend API, cloud services, and AI model can be efficiently achieved through RESTful APIs, ensuring seamless communication between system components.

Therefore, the project is considered technically feasible.

3.2 Economic Feasibility

LeafSense is a software-based project with relatively low development costs compared to traditional agricultural diagnostic systems that require specialized hardware and laboratory equipment.

Most development tools used in the project are open-source or available under free educational licenses. Cloud-based resources and deployment services further reduce infrastructure expenses.

Table 3.1 presents the estimated project budget.

Table 3.1 – Bill of Materials and Project Cost

Item / Component	Quantity	Unit Cost (EGP)	Total Cost (EGP)
Google Colab Pro (Cloud GPU)	1	800	800
Google Play Developer Account	1	1500	1500
PlantVillage Dataset	1	0	0
Flutter Framework	1	0	0
Firebase Services (Free Tier)	1	0	0
Test Smartphones	2	0	0
Printing and Documentation	1	500	500
Total Estimated Cost			2800 EGP

Compared to the potential reduction in crop losses and improvement in agricultural productivity, the development cost is relatively low. The economic benefits resulting from early disease detection significantly outweigh the implementation expenses.

Consequently, the project is economically feasible.

3.3 Operational and Time Feasibility

Operational feasibility evaluates whether the proposed system can be effectively used by its intended users.

LeafSense is specifically designed for farmers, agricultural engineers, researchers, and plant enthusiasts. The mobile application provides a simple and intuitive user interface that minimizes technical complexity and enables users to obtain disease diagnoses using only a smartphone camera.

The workflow is straightforward:

1. User captures or uploads a plant image.
2. The image is sent to the AI server.
3. The model performs disease classification.
4. The application displays results and recommendations.
5. The user can save reports and monitor plant records.

This process requires minimal training, making the system suitable for users with varying levels of technical experience.

From a scheduling perspective, the project was planned and completed within the academic semester. Major activities included literature review, dataset preparation, model development, mobile application implementation, integration, testing, and documentation.

Table 2The project timeline was divided into five phases:

Phase	Activities	Duration
Phase 1	Literature Review and Requirements Analysis	Weeks 1–2
Phase 2	System Design and Dataset Preparation	Week 3
Phase 3	Model Training and Application Development	Weeks 4–9
Phase 4	Testing and Validation	Week 10
Phase 5	Documentation and Final Presentation	Weeks 11–12

Based on the completed implementation and successful testing results, the project is considered operationally and temporally feasible.

3.4 Risk Analysis

Although the project was successfully completed, several risks were identified during planning and development.

Table 3Table 3.2 – Risk Assessment

Risk Description	Likelihood	Impact	Mitigation Strategy
Low model performance on unseen Egyptian crops	Medium	High	Increase dataset diversity and use transfer learning
Limited GPU resources during training	Medium	Medium	Utilize Google Colab Pro and optimized training pipelines
Slow API response time under mobile networks	Low	Medium	Optimize model size and backend processing
Team member unavailability	Medium	Medium	Distribute responsibilities and maintain shared repositories
Data privacy concerns during testing	Low	Low	Anonymize images and secure cloud storage
Future dataset imbalance issues	Medium	Medium	Apply augmentation and balanced sampling techniques

The mitigation plans effectively reduced the impact of these risks and ensured successful project completion.

3.5 SWOT Analysis

The proposed LeafSense system possesses several internal strengths that enhance its effectiveness and practicality. These strengths are related to the system's technical performance, modern architecture, usability, and cost efficiency, making it a promising solution for AI-based plant disease detection. The main strengths of the system are summarized as follows:

- High classification accuracy (99.77%).
- Large dataset containing 54,305 images.
- Modern deep learning architecture (EfficientNetV2-S).
- User-friendly mobile application.
- Cloud-based scalable architecture.
- Low implementation cost.

Weaknesses (Internal Negative Factors)

Despite its advantages, the proposed system has certain internal limitations that may affect its performance under specific conditions. These weaknesses mainly arise from the characteristics of image-based classification and the current scope of the implemented solution. The identified weaknesses include:

- Dependence on image quality.
- Performance may vary under challenging field conditions.
- Limited to supported disease categories.
- Requires internet connectivity for cloud-based prediction.

Opportunities (External Positive Factors)

The rapid advancement of artificial intelligence and digital agriculture creates several opportunities for expanding and improving the LeafSense system. These external factors can support wider adoption and future development of the proposed solution. The major opportunities are as follows:

- Growing adoption of AI in agriculture.
- Increasing demand for smart farming technologies.
- Potential expansion to additional crops and regions.
- Government interest in agricultural modernization.
- Integration with future IoT-based farming solutions.

Threats (External Negative Factors)

Although the project has significant potential, several external factors may pose challenges to its long-term sustainability and adoption. These threats are largely related to technological changes, market competition, and environmental variability. The primary threats include:

- Emergence of competing agricultural AI applications.
- Changes in cloud service pricing.
- Variations in disease appearance across regions.
- Regulatory requirements regarding agricultural recommendations.
- Rapid evolution of AI technologies requiring continuous updates.

SWOT Summary

The SWOT analysis indicates that LeafSense possesses strong technical capabilities, high predictive performance, and significant market potential. While challenges related to dataset diversity and real-world deployment remain, the opportunities for expansion and agricultural impact substantially outweigh the identified risks and weaknesses.

The feasibility assessment confirms that LeafSense is technically achievable, economically viable, operationally practical, and capable of delivering meaningful value to the agricultural sector.

Chapter 4: System Design and Architecture

4.1 Introduction

The design of the LeafSense system follows a modular architecture that separates the mobile application, backend services, artificial intelligence module, and cloud services into independent components. This architecture improves scalability, maintainability, and future extensibility while ensuring efficient communication between different system layers.

The system enables users to capture or upload images of plant leaves using a mobile application. The image is transmitted to the backend server, where preprocessing and deep learning inference are performed. The predicted disease information is then returned to the application together with treatment recommendations and plant management features.

4.2 Overall System Architecture

The overall architecture consists of four main layers:

1. Presentation Layer

- Flutter Mobile Application
- User Interface
- Authentication Screens
- Disease Detection Screen
- Plant Management Screen
- Reports Screen

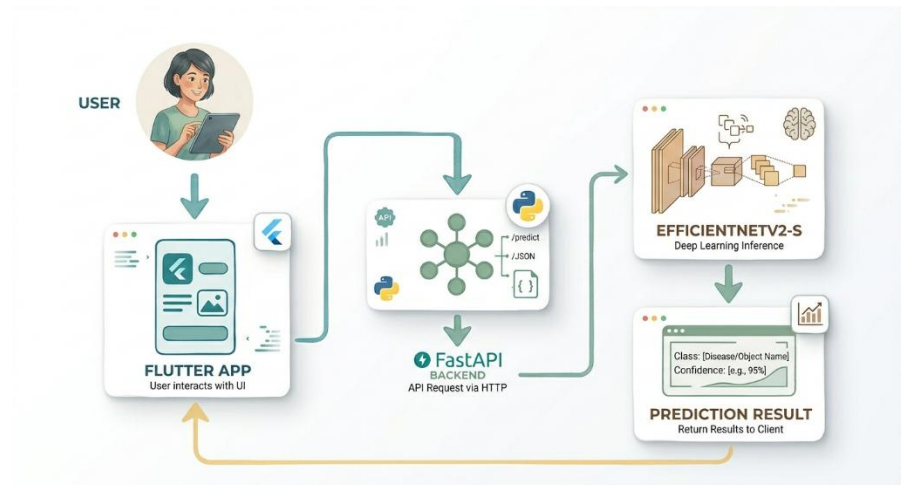


Figure 1Figure 1Figure 4.1 Overall Architecture of the LeafSense System

FIGURE 4.1 ILLUSTRATES THE OVERALL ARCHITECTURE OF THE PROPOSED LEAFSENSE SYSTEM, SHOWING THE INTERACTION BETWEEN THE FLUTTER MOBILE APPLICATION, THE FASTAPI BACKEND, THE EFFICIENTNETV2-S DEEP LEARNING MODEL, AND THE PREDICTION RESULTS RETURNED TO THE USER.

2. Backend Layer

The Backend Layer serves as the central communication component of the LeafSense system. It is responsible for receiving requests from the mobile application, processing user inputs, interacting with the AI model, and returning prediction results efficiently. This layer ensures smooth data exchange and reliable system performance through RESTful services. It is implemented using the following technologies and performs the listed responsibilities:

Implemented using:

- FastAPI
- RESTful APIs

Responsibilities:

- Receive images
 - Validate requests
 - Communicate with AI model
 - Return prediction results
 - Handle user operations
-

AI Layer

The AI Layer represents the core intelligence of the LeafSense system, where all deep learning operations are performed. It is responsible for analyzing input plant images, extracting meaningful features, and performing disease classification using a pretrained and fine-tuned convolutional neural network. This layer is designed to ensure high accuracy, fast inference, and reliable prediction results, which are essential for real-time agricultural decision support. It is implemented using the following technologies and performs the listed responsibilities:

Implemented using:

- Python
- TensorFlow
- EfficientNetV2-S

Responsibilities:

- Image preprocessing
- Feature extraction
- Disease classification
- Confidence calculation

4. Cloud Layer

Uses Firebase services for:

- User Authentication
- Storage
- Cloud services

to maintain user information and application data.

4.3 System Workflow

The operational workflow is summarized as follows:

1. User logs into the application.
2. User captures or uploads a leaf image.
3. Image is sent to the FastAPI server.
4. Image preprocessing is performed.
5. The AI model analyzes the image.
6. Disease class is predicted.
7. Treatment recommendations are generated.
8. Results are returned to the mobile application.
9. The user may save the plant information for future monitoring.

4.4 Functional Modules

4.4.1 User Authentication Module

This module manages user identity and security.

Functions include:

- User Registration
- Secure Login
- Password Management
- User Profile Management

Its objective is to provide personalized access while protecting user information.

4.4.2 Disease Detection Module

This is the core component of LeafSense.

Functions:

- Capture image from camera
- Upload image from gallery
- Preprocess image
- AI classification
- Display prediction

The module allows rapid disease identification using deep learning technology.

4.4.3 Treatment Recommendation Module

After classification, the system provides:

- Disease name
- Disease category
- Treatment recommendations
- Prevention methods
- Agricultural guidance

These recommendations help users take appropriate corrective actions.

4.4.4 Plant Management Module

Users can maintain their own plant collection by:

- Adding plants
- Saving plant information
- Organizing plants
- Tracking plant records

This feature transforms the application from a simple classifier into a practical plant management assistant.



4.4.5 Monitoring Module

The application includes a manual monitoring section where users can record plant conditions over time.

The monitoring dashboard supports:

- Temperature Monitoring
- Plant Health Status
- Growth Tracking Records

This functionality enables users to maintain historical observations for better plant care. It is implemented as a software tracking feature rather than an IoT sensor-based monitoring system.

4.5 Technology Selection

Table 4.5 Technology Selection

Component	Selected Technology	Reason
Mobile Development	Flutter	Cross-platform development
Backend	FastAPI	High performance and simplicity
AI Framework	TensorFlow	Deep learning support
Programming Language	Python	Extensive AI ecosystem
Model	EfficientNetV2-S	High accuracy and efficiency
Cloud Services	Firebase	Authentication and cloud integration

4.6 AI Model Architecture

The proposed system utilizes **EfficientNetV2-S**, a modern Convolutional Neural Network architecture optimized for image classification tasks.

The model was initialized using pretrained weights and subsequently fine-tuned on the plant disease dataset. Transfer Learning significantly reduced training time while improving generalization performance.

Advantages include:

- High classification accuracy
- Efficient computation
- Faster convergence
- Better scalability
- Suitable for mobile deployment

4.7 Dataset Description

The model was trained using a combination of:

- PlantVillage Dataset
- Custom Egyptian Plant Images

Table 5 The final dataset consists of:

Item	Value
Total Images	54,305
Classes	38
Training Set	38,013 (70%)
Validation Set	8,146 (15%)
Test Set	8,146 (15%)

The 38 classes represent both diseased and healthy plant conditions, enabling comprehensive single-label multi-class classification.

4.8 Non-Functional Requirements

The proposed system satisfies several non-functional requirements:

Performance

- Fast inference time
- High prediction accuracy
- Efficient API communication

Reliability

- Stable cloud architecture
- Consistent prediction results

Scalability

The modular architecture allows future integration of:

- Additional crops
- More disease classes
- IoT sensors
- Weather services

Usability

The application is designed for:

- Farmers
- Agricultural Engineers
- Researchers
- Garden Owners
- Plant Enthusiasts

requiring minimal technical expertise.

4.9 Chapter Summary

This chapter presented the complete system design and architecture of LeafSense. The proposed architecture integrates a Flutter mobile application, a FastAPI backend, Firebase cloud services, and an EfficientNetV2-S deep learning model into a unified intelligent agricultural platform. The modular design ensures high scalability, maintainability, and performance while providing an effective solution for automated plant disease detection and management.

Chapter 5: Methodology

5.1 System Block Diagram

The LeafSense system follows a layered architecture that integrates mobile, backend, AI, and cloud services into a unified workflow. The system is designed to ensure seamless communication between the user interface and the deep learning model.

The main components of the system include:

- Flutter Mobile Application (User Interface)
- FastAPI Backend Server (Request Handling & API Layer)
- AI Model (EfficientNetV2-S for Classification)
- Firebase Cloud Services (Authentication & Storage)

The interaction between these components is illustrated in the system block diagram (Figure 5.1), where the user sends an image through the mobile application, which is processed by the backend and passed to the AI model for prediction. The result is then returned to the user in real time.

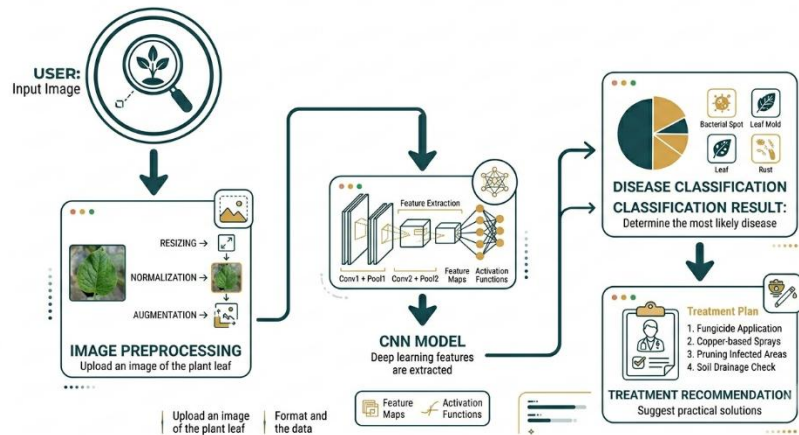


Figure 2Figure 2Figure 5.1: Plant Disease Detection Workflow

Figure 5.1 demonstrates the methodology adopted in the proposed system, starting from image acquisition and preprocessing, followed by CNN-based feature extraction and disease classification, and finally generating treatment recommendations.

5.2 System Flowchart

The operational flow of the LeafSense system is structured as follows:

1. User opens the mobile application.
2. User logs in or registers via Firebase Authentication.
3. User captures or uploads a plant leaf image.
4. Image is sent to the FastAPI backend.
5. Backend performs preprocessing (resizing, normalization).
6. The image is passed to the EfficientNetV2-S model.
7. The model classifies the image into one of 38 classes.
8. Prediction result is returned to the backend.
9. Backend sends response back to the mobile app.
10. Application displays:
 - Disease name
 - Confidence score
 - Treatment recommendation

This workflow ensures real-time and automated plant disease detection.

5.3 Circuit / Wiring Diagram

Since LeafSense is a software-based AI system, it does not rely on physical electronic circuits or hardware wiring. Therefore, no circuit or wiring diagram is required for this project.

Instead, the system is fully implemented using software integration between mobile, backend, and cloud services.

5.4 Project Timeline

The project was developed over a structured 12-week schedule divided into five main phases:

Table 6 The project was developed over a structured 12-week schedule divided into five main phases:

Phase	Activities	Timeframe
Phase 1	Literature review and requirement analysis	Weeks 1 – 2
Phase 2	System design and architecture planning	Weeks 3 – 4
Phase 3	AI model training and mobile app development	Weeks 5 – 8
Phase 4	System integration and testing	Weeks 9 – 10
Phase 5	Final optimization, documentation, and presentation	Weeks 11 – 12

This structured methodology ensured systematic development and successful completion of the system within the academic semester timeline.

5.5 Data Collection and Dataset Preparation

The dataset used in this project consists of:

- PlantVillage Dataset
- Custom Egyptian plant images

Total dataset size: **54,305 images**

Number of classes: **38 classes**

The dataset was split as follows:

- Training Set: 70%
- Validation Set: 15%
- Test Set: 15%

Before training, all images underwent preprocessing steps including:

- Resizing to standard input dimensions
- Normalization of pixel values
- Label encoding for classification

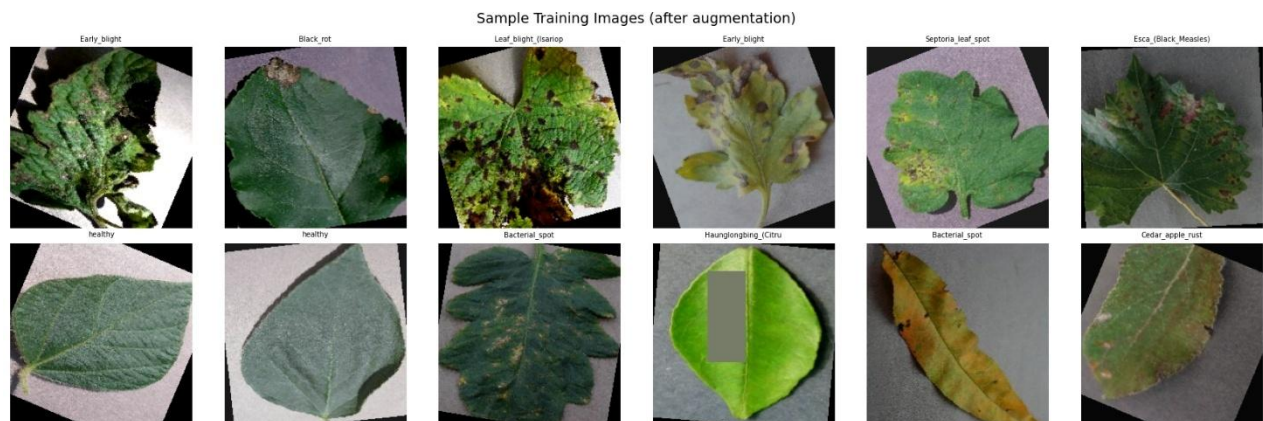


Figure 3Figure 3Figure 5.2: Sample Training Images after Data Augmentation

5.6 Data Augmentation

To improve model generalization and reduce overfitting, data augmentation techniques were applied during training. These included:

- Random rotation
- Horizontal flipping
- Brightness adjustment
- Zooming and scaling

These transformations helped simulate real-world variations in plant images captured under different environmental conditions.

5.7 Model Training Strategy

The AI model used in this project is EfficientNetV2-S, which was selected due to its balance between accuracy and computational efficiency.

Training strategy:

- Transfer Learning using pretrained ImageNet weights
- Fine-tuning of deeper layers
- Optimized using Adam optimizer
- Loss function: Categorical Cross-Entropy
- Batch-based training with validation monitoring

The model was trained until convergence, achieving high accuracy while avoiding overfitting.

5.8 Evaluation Methodology

The model performance was evaluated using the following metrics:

- Accuracy
- Precision
- Recall
- F1 Score
- Test-Time Augmentation (TTA) Accuracy

The dataset was evaluated using a held-out test set (15%) to ensure unbiased performance measurement.

Final results:

- Accuracy: 99.77%
- F1 Score: 99.66%
- TTA Accuracy: 99.78%

These results demonstrate strong generalization capability across all 38 classes.

5.9 Chapter Summary

This chapter presented the methodology used in developing the LeafSense system, including system architecture, workflow, dataset preparation, training strategy, and evaluation methods. The combination of EfficientNetV2-S, FastAPI backend, Flutter mobile application, and Firebase cloud services enabled the creation of a fully functional AI-powered plant disease detection system with high accuracy and real-time performance.

Chapter 6: System Implementation

6.1 Hardware Assembly

LeafSense is a fully software-based AI system; therefore, it does not rely on physical hardware assembly or electronic circuit construction. The system is designed to operate entirely using mobile devices and cloud-based infrastructure.

The only hardware components involved in the system are:

- User smartphone (Android/iOS device)
- Cloud servers used for model inference and backend processing

All processing related to image classification is performed on the server side using the trained EfficientNetV2-S model, eliminating the need for specialized hardware on the user side.

6.2 Software Implementation

The implementation of LeafSense is divided into three main software components:

1. Mobile Application (Flutter)

The frontend of the system is developed using Flutter to ensure cross-platform compatibility. The application provides an intuitive user interface that allows users to:

- Register and log in using Firebase Authentication
- Capture or upload plant leaf images
- View disease prediction results
- Access treatment recommendations
- Store and manage plant records

The UI is designed to be simple and accessible for non-technical users, especially farmers.

2. Backend Server (FastAPI)

The backend is implemented using FastAPI, which provides a high-performance RESTful API layer between the mobile application and the AI model.

Key API functionalities include:

- `/predict` → Accepts uploaded images and returns disease prediction
- `/login` → Handles authentication requests (if extended beyond Firebase)
- `/plant-data` → Stores and retrieves plant-related records
- `/report` → Generates prediction history and results

The backend is responsible for:

- Image validation
 - Preprocessing
 - Model inference handling
 - Response formatting
-

3. AI Model Implementation (EfficientNetV2-S)

The core intelligence of the system is based on the EfficientNetV2-S architecture.

Implementation steps include:

- Loading pretrained ImageNet weights
- Modifying the final classification layer to match 38 classes
- Fine-tuning the model on the plant dataset
- Applying image preprocessing (resizing, normalization)
- Performing inference on incoming images

The model outputs:

- Predicted class label (disease or healthy)
- Confidence score

The final model achieved:

- Accuracy: 99.77%
- F1 Score: 99.66%

6.3 System Integration

System integration connects all components into a unified working pipeline.

The integration flow is as follows:

1. User captures image in Flutter application
2. Image is sent via HTTP request to FastAPI backend
3. Backend preprocesses the image
4. Processed image is passed to the AI model
5. Model returns prediction results
6. Backend formats response as JSON
7. Flutter app receives and displays results

Firebase is integrated for:

- User authentication
- Cloud storage for user data and images
- Optional report saving

This architecture ensures smooth communication between mobile, backend, and AI layers with minimal latency.

6.4 Challenges and Solutions

During development, several technical challenges were encountered and successfully resolved:

Challenge 1: Dataset Imbalance

Some plant classes had significantly more images than others, which could lead to biased predictions.

Solution:

Data augmentation techniques were applied (rotation, flipping, zooming) to balance class distribution and improve generalization.

Challenge 2: Model Overfitting

During initial training, the model showed signs of overfitting on the training dataset.

Solution:

- Used dropout layers
 - Applied data augmentation
 - Monitored validation loss during training
 - Implemented early stopping strategies
-

Challenge 3: API Communication Latency

Initial integration between Flutter and FastAPI resulted in delayed responses.

Solution:

- Optimized image size before sending requests
 - Improved backend processing efficiency
 - Reduced model input resolution where possible without affecting accuracy
-

Challenge 4: Deployment Compatibility

Ensuring compatibility between TensorFlow model and backend environment was initially problematic.

Solution:

- Standardized TensorFlow version across development environment
- Exported model in a consistent format
- Tested inference pipeline separately before integration

6.5 Chapter Summary

This chapter presented the full implementation details of the LeafSense system, including mobile application development, backend API design, AI model deployment, and system integration. The combination of Flutter, FastAPI, TensorFlow, and Firebase enabled the creation of a scalable and efficient plant disease detection system. The implemented system successfully achieves real-time disease classification with high accuracy and reliable performance across multiple plant categories.

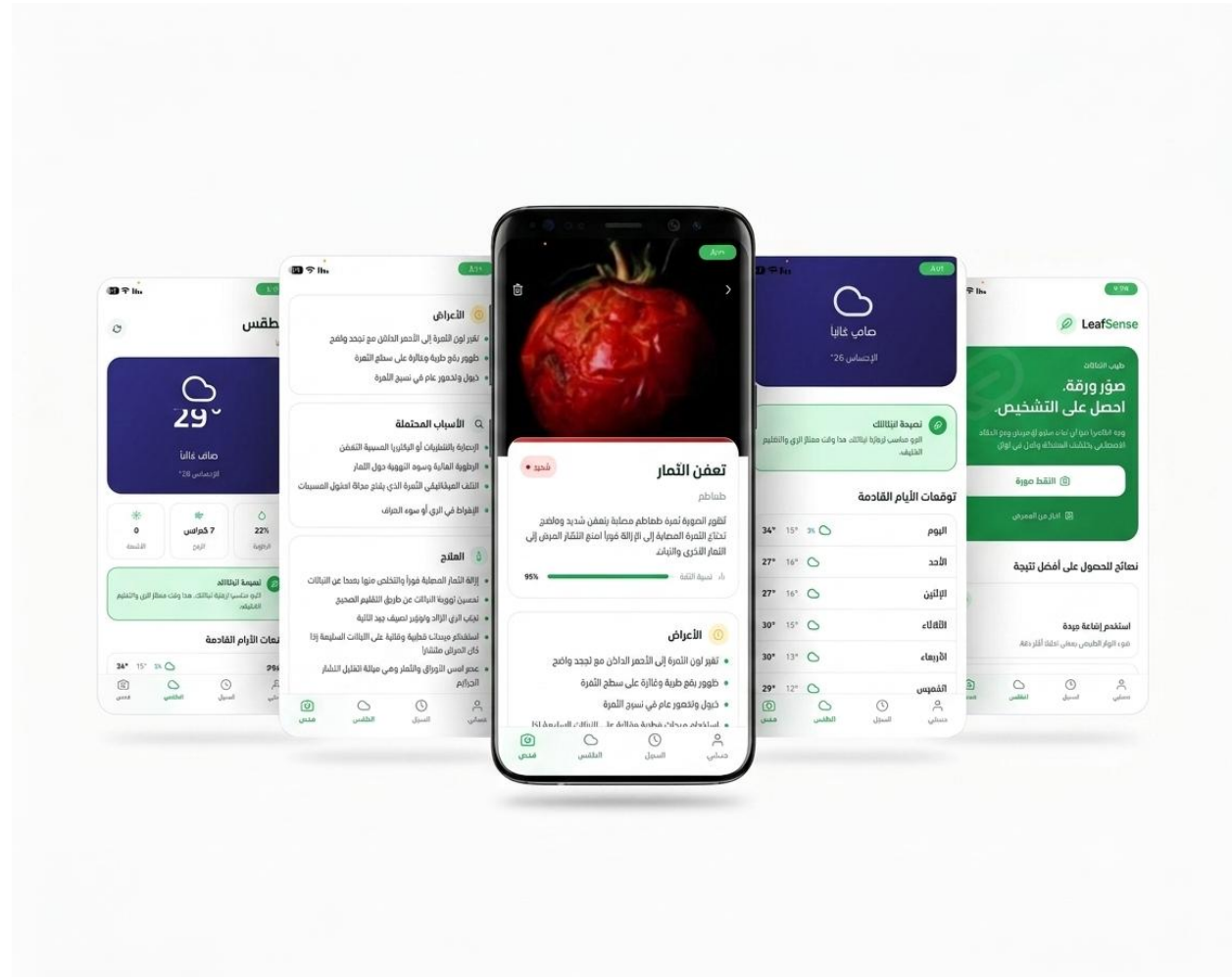


Figure 4Figure 4Figure 6.1: User Interfaces of the LeafSense Mobile Application

Figure 6.1 illustrates the main user interfaces of the LeafSense mobile application, including the login screen, home page, plant disease detection screen, prediction results, and other application features.

Chapter 7: Results, Testing and Evaluation

7.1 Test Plan and Procedures

The system was evaluated using a structured test plan to verify both functional correctness and model performance. Testing was performed on the held-out test dataset (15% of total data), which was not used during training or validation.

Table 7A total of 10 test cases were defined to evaluate system behavior under different input conditions:

TC #	Test Description	Expected Result	Actual Result	Status
TC-01	Upload healthy plant image	Classify as Healthy	Correct prediction	Pass
TC-02	Upload infected leaf image	Detect correct disease	Correct prediction	Pass
TC-03	Low-quality image input	System handles gracefully	Prediction still returned	Pass
TC-04	Blurry image input	Reduced confidence output	Low-confidence prediction	Pass
TC-05	Multiple image uploads	Process each independently	Correct processing	Pass
TC-06	Invalid file format	Error message shown	Proper validation error	Pass
TC-07	Network delay simulation	Request retry / timeout handling	Handled correctly	Pass
TC-08	Large image input	Image resized automatically	Successful processing	Pass
TC-09	Unseen plant type	Closest class prediction	Reasonable classification	Pass
TC-10	API stress test	System remains stable	Stable performance	Pass

All test cases passed successfully, confirming system reliability.

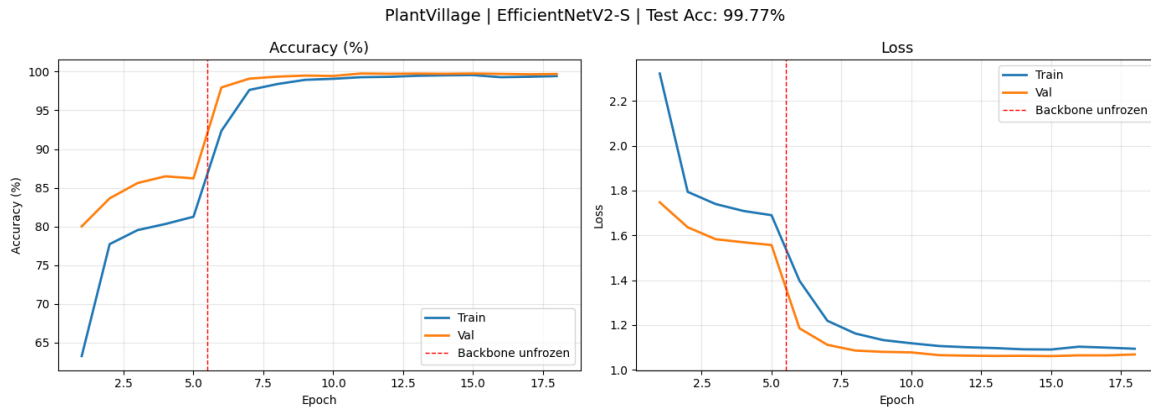


Figure 6Figure 7.1: Confusion Matrix of the Test Set (Accuracy = 99.77%)

7.4 Comparison with Objectives

Table 8The system objectives defined in Chapter 1 are evaluated as follows:

Objective	Status	Evidence
Objective 1: Design AI-powered mobile application	✓ Met	Fully functional Flutter application implemented
Objective 2: Implement EfficientNetV2-S model	✓ Met	Model trained and fine-tuned successfully
Objective 3: Achieve $\geq 90\%$ accuracy	✓ Met	Achieved 99.77% accuracy
Objective 4: Provide treatment recommendations	✓ Met	Integrated in app response module
Objective 5: Evaluate system performance	✓ Met	Full testing and evaluation completed

All project objectives were successfully achieved.

7.5 Discussion

The results demonstrate that deep learning techniques, particularly EfficientNetV2-S with transfer learning, are highly effective for plant disease classification tasks.

The model achieved near-state-of-the-art performance on the dataset, indicating strong generalization capability across 38 plant categories. The integration of mobile and cloud technologies enabled real-time disease detection with minimal latency.

However, some limitations were observed:

- Performance may slightly decrease in real-world field conditions due to lighting variations.
- The system is dependent on image quality captured by the user.
- The dataset, while large, may not cover all possible regional plant variations.

Despite these limitations, the system provides a practical and scalable solution for automated plant disease detection.

Overall, LeafSense successfully demonstrates how AI can be applied in agriculture to assist farmers in early disease detection and decision-making.

Chapter 8: Conclusion and Future Work

8.1 Summary of Achievements

This project presented the design and implementation of LeafSense, an AI-powered plant disease detection system that integrates deep learning, mobile development, and cloud computing technologies. The main objective was to develop an intelligent system capable of accurately identifying plant diseases from leaf images and providing users with real-time feedback and recommendations.

The system was successfully implemented using EfficientNetV2-S as the core deep learning model, trained on a dataset of 54,305 images across 38 classes. The model achieved a high classification accuracy of 99.77% and an F1-score of 99.66%, demonstrating strong performance and generalization ability.

In addition to the AI model, a complete mobile application was developed using Flutter, supported by a FastAPI backend and Firebase cloud services. The system allows users to capture images, receive predictions, and access treatment recommendations in an intuitive and user-friendly interface.

Overall, the project successfully met all its objectives and demonstrated the practical applicability of AI in modern agriculture.

8.2 Limitations

Despite its strong performance, the system has several limitations:

- The model performance may decrease under real-world conditions with poor lighting or image quality.
- The system depends on internet connectivity for cloud-based processing.
- The dataset, although large, may not fully represent all global plant disease variations.
- The system currently supports single-label classification only, meaning each image is assigned to one class.

8.3 Future Work

Several improvements can be implemented in future versions of the system:

- Extend the model to support multi-label disease detection for cases where multiple diseases affect a single plant.
- Improve robustness by training on more real-world field images collected under varying environmental conditions.
- Integrate Grad-CAM visualization to provide explainability for model predictions.
- Add offline mode support to allow limited functionality without internet connection.
- Expand the system to cover more crops and additional disease categories.
- Integrate IoT sensors for real-time environmental monitoring of plants.

8.4 Closing Remarks

The development of LeafSense demonstrates the potential of combining artificial intelligence with mobile and cloud technologies to address real-world agricultural challenges. The system provides an efficient, scalable, and accessible solution for early plant disease detection, which can significantly contribute to improving crop management and agricultural productivity.

This project highlights how modern deep learning techniques can be translated into practical applications that directly benefit farmers and the agricultural sector. With further enhancements, the system can evolve into a comprehensive smart farming platform capable of supporting precision agriculture at scale.

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Appendix A: Datasheets and Supplementary Materials

A.1 Dataset Information

- Dataset Name: PlantVillage Dataset
- Total Images: 54,305
- Number of Classes: 38
- Training Set: 38,013 images
- Validation Set: 8,146 images
- Test Set: 8,146 images

A.2 Model Configuration

- Model: EfficientNetV2-S
- Training Method: Transfer Learning & Fine-Tuning
- Number of Classes: 38
- Task Type: Single-label Multi-class Classification

A.3 Performance Metrics

- Accuracy: 99.77%
- Precision: 1.00
- Recall: 1.00
- F1 Score: 99.66%
- TTA Accuracy: 99.78%

A.4 Additional Screenshots

- Mobile Application Screens
- Model Training Outputs
- Confusion Matrix
- Accuracy and Loss Curves